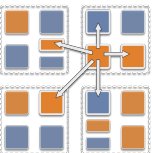


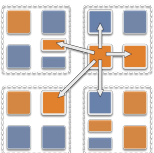
Live visualization

- A tool to view the execution of Charm++ programs that produce a 2D image
 - such as the particle exercise
- Each chare “deposits” information for its particles
 - Coordinates
 - Color
 - Size
- Aggregates into one image



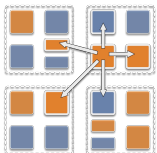
Why use liveViz?

- Demonstrates the load-balancing capabilities of Charm4Py
- Experiment with different load balancers, show how particles behave in each case
 - GreedyLB
 - RefineLB
 - Any other load balancer supported by TreeLB

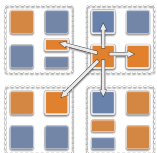
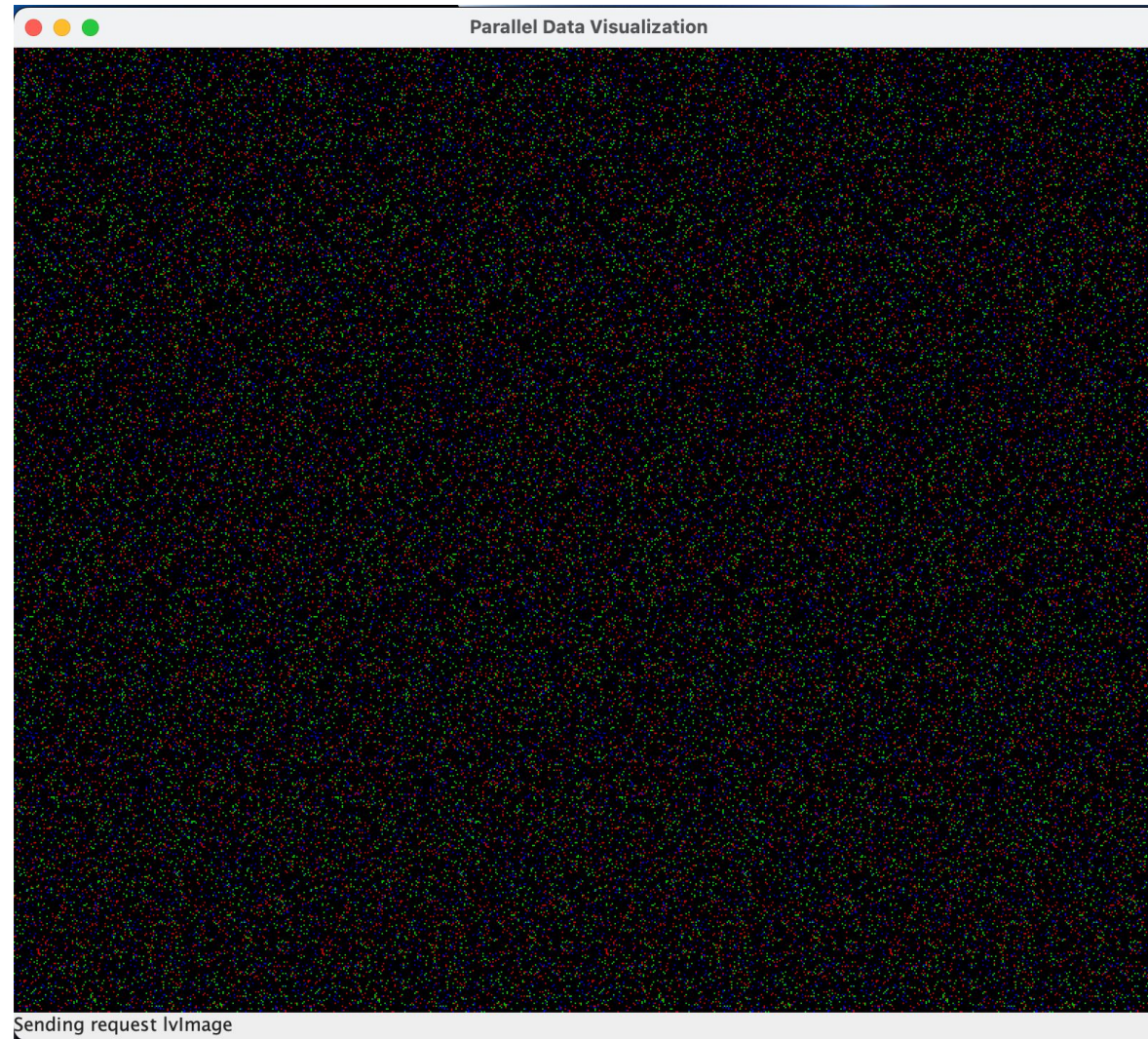


LiveViz and CCS

- Converse client-server (CCS) uses TCP to communicate between Charm application and external application
 - Charm4Py program as server
 - Java application as client
- LiveViz communicates with running particle program to display image of particles

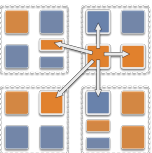


Example output



Download and install

- liveViz client
 - git clone https://github.com/UIUC-PPL/ccs_tools.git
 - cd ccs_tools
 - ant
- Charm liveViz build
 - go to Charm++ install directory
 - cd src/libs/ck-lib/liveViz
 - make

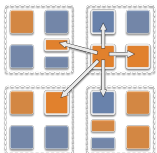


Initializing liveViz in code

- Create a config, then call init

```
def main(args):  
    a = Array(Class, dims=(16,16))  
    config = liveviz.Config()  
    liveviz.LiveViz.init(config, a.makeImg)
```

- Config parameters:
 - isColor=True: Defines whether image is RGB or black and white
 - isPush=True: Constantly request for images when True, only request on window resize when false



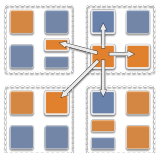
Instrumenting code for liveViz

- Start by defining an entry method within a chore array with a “request” argument
 - request: a request for image data

```
def makeImg(self, request)
```

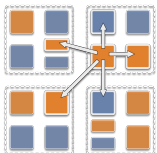
- In this method, do work to set up the image, then call `liveviz.LiveViz.deposit`

```
liveviz.LiveViz.deposit(data, self, start_x, start_y, local_height,  
local_width, global_height, global_width)
```



Instrumenting code for liveViz

- Arguments to deposit function
 - data: Row-major byte array of pixels, 1 byte/pixel when black and white, 3 byte/pixel when RGB
 - start_x, start_y: Coordinates of top left corner of my chare's section
 - local_height, local_width: Height and width (in pixels) of my chare's section (typically the same for all chares)
 - global_height, global_width: Height and width of the entire image



Running liveViz

Server (particle program)

```
$ cd  
charm4py/examples/liveviz  
$ python -m charmrun.start  
+p4 ++server ++local  
liveviz.py ++server-port 1234
```

Client

```
$ ~/ccs_tools/bin/liveViz  
localhost 1234
```